

INTERNOVA — WEEK 6

FINAL DATA ANALYTICS PROJECT

Superstore Sales & Profitability Analysis

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InternNova Internship — Week 6

11 September 2026

1. Project Overview

This final Week 6 project brings together data preparation, exploratory data analysis, Python visualization, Power BI business intelligence, business recommendations, and Git/GitHub workflow.

The objective is to transform Superstore sales data into actionable insights covering sales performance, profitability, category and regional performance, monthly trends, and discount impact.

Problem Statement: Transform sales data into actionable insights that identify strong and weak product areas, regional performance differences, profitability concerns, and trends over time.

Tools: Python (Pandas, Matplotlib, Seaborn), Power BI, DAX, Git, GitHub, and Microsoft PowerPoint.

2. Dataset Selection & Data Preparation

Dataset: superstore_sales_analysis_cleaned.csv.

The dataset contains 10,194 records and 21 columns. Validation confirmed 0 missing values and 0 duplicate rows, so no additional record-level correction was required.

Key analytical fields include Sales, Quantity, Discount, Profit, Category, Sub-Category, Region, and Order Date.

Check	Result
Records	10,194
Columns	21
Missing values	0
Duplicate rows	0

3. Exploratory Data Analysis (EDA)

Descriptive statistics: Sales count = 10,194; mean Sales = 228.2256; median Sales = 53.91; minimum Sales = 0.444; maximum Sales = 22,638.48.

Positive-profit transactions: 8,228, representing approximately 80.71% of transactions.

Correlation: Sales–Profit ≈ 0.48 ; Sales–Quantity ≈ 0.20 ; Discount–Profit ≈ -0.22 ; Quantity–Profit ≈ 0.07 .

Key findings: Technology leads sales; West is the highest-sales region; Tables has negative total profit; higher discount levels are associated with weaker profitability; monthly sales peak around December 2025.

4. Data Visualizations

Five Python visualizations were created: Sales by Product Category; Profit by Product Category; Sales by Region; Monthly Sales Trend; Discount vs Profit.

These visuals communicate category comparisons, profitability, regional performance, time trends, and the discount-profit relationship.

5. Power BI Dashboard

A two-page Power BI report was developed with KPI cards for Total Orders, Total Profit and Total Sales; Sales by Region; Sales by Category; Monthly Sales Trend; Category & Sub-Category Performance; Region and Category slicers; and a Business Insights & Recommendations page.

Core DAX measures: Total Sales = SUM(Sales); Total Profit = SUM(Profit); Total Orders = DISTINCTCOUNT(Order ID); Average Order Value = Total Sales / Total Orders; Profit Margin % = Total Profit / Total Sales.

A star-schema approach was used with DimCustomer, DimProduct and DimDate dimension tables connected to the sales fact table.

POWER BI DASHBOARD — SCREENSHOT

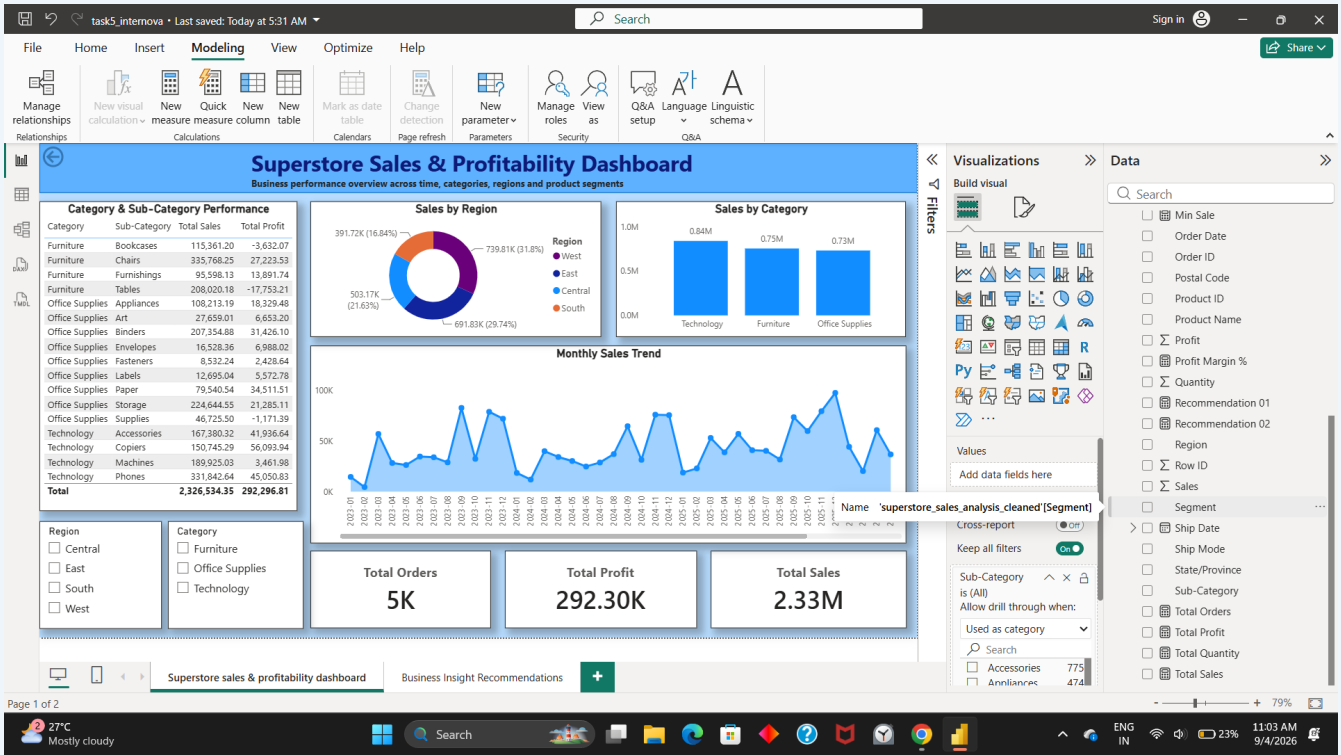


Figure 1. Main Superstore Sales & Profitability Dashboard.

6. Business Insights & Recommendations

- Insight 1: Technology leads category sales at approximately 0.84M.
- Insight 2: Office Supplies has the lowest category sales at approximately 0.73M.
- Insight 3: Tables has approximately -17.75K total profit and is the key identified profitability concern.
- Insight 4: West is the highest-sales region at approximately 0.74M.
- Insight 5: Monthly sales reach their highest observed point around December 2025 at approximately 98K.
- Recommendation 1: Prioritize Technology products and maintain availability of strong-performing products while monitoring profitability.
- Recommendation 2: Review pricing and discount strategy for Tables to reduce negative-margin transactions and improve sub-category profitability.

BUSINESS INSIGHTS & RECOMMENDATIONS — SCREENSHOT

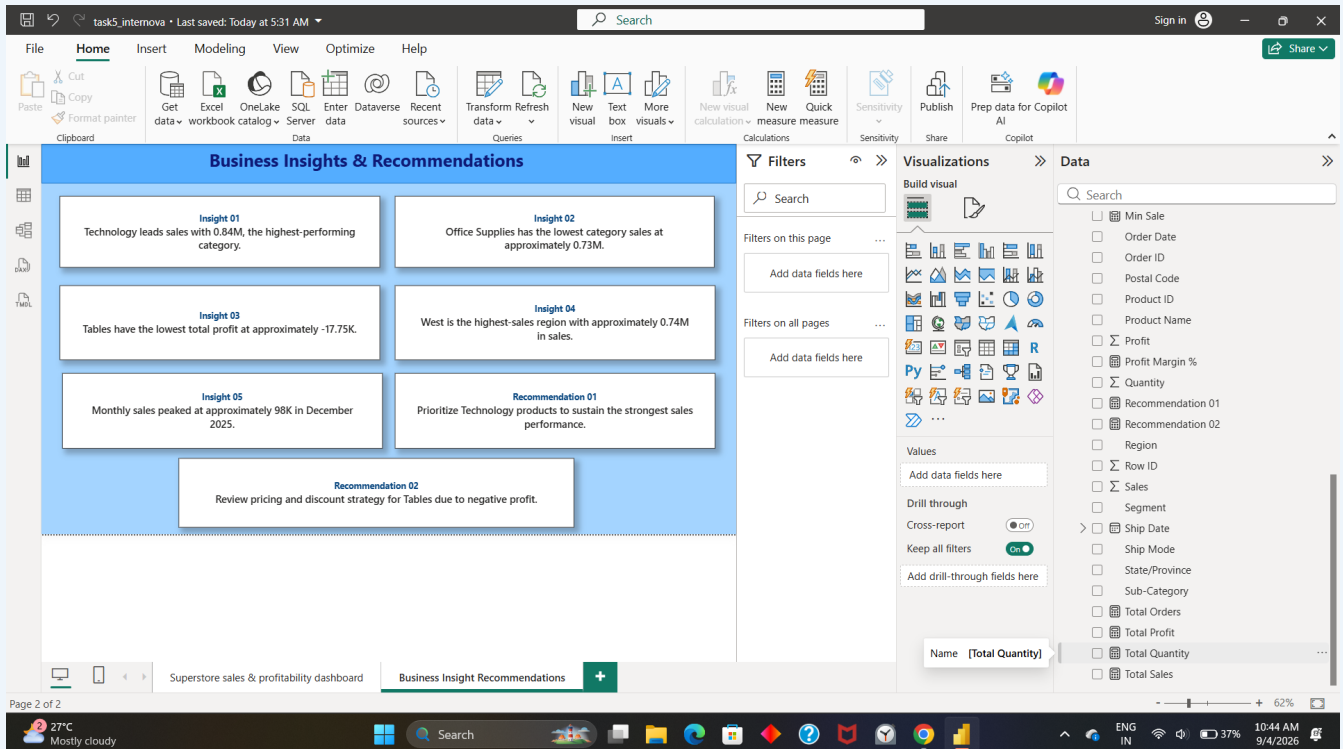


Figure 2. Power BI insights and recommendations page.

7. Git & GitHub Workflow

Git was used for version control and GitHub for the remote repository. Meaningful commits covered repository setup, the cleaned dataset, EDA, visualizations, Power BI dashboard, business insights/recommendations, and the final presentation.

The final presentation was staged, committed, and successfully pushed to the main branch.

8. Repository Structure

- Dataset/ — prepared dataset.
- EDA/ — exploratory analysis and documentation.
- Visualizations/ — Python scripts and generated charts.
- PowerBI/ — Power BI dashboard file.
- Insights_Recommendations.md — business findings.
- README.md — project documentation.
- InternNova_Week6_Final_Project_Presentation_Nandani_Rana.pptx — final presentation.

9. Conclusion

This project demonstrates an end-to-end analytics workflow from data validation and EDA through visualization, Power BI reporting, business recommendations, and Git/GitHub delivery.

The analysis identifies Technology and the West region as strong sales areas, while Tables is a notable profitability concern. The dashboard and recommendations translate these findings into practical business actions.